

A short note on Variance Swaps

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Abstract

A short note on Variance Swaps and volatility derivatives

Replicating a twice-differentiable payoff with a static position in vanilla options

Any twice differentiable payoff $g(S_T)$ at time T may be replicated using a static portfolio of European options expiring at time T .

From the Breeden-Litzenberger formula, we know that we may write the *pdf* of the stock-price S_T as:

$$p(S_T, T; S_t, t) = \left. \frac{\partial^2 \tilde{C}(K, T, S_t, t)}{\partial K^2} \right|_{K=S_T} = \left. \frac{\partial^2 \tilde{P}(K, T, S_t, t)}{\partial K^2} \right|_{K=S_T} \quad (1)$$

where $\tilde{C}$ and $\tilde{P}$ represent the undiscounted call and put prices respectively.

Then, the value of a claim with a generalized payoff $g(S_T)$ at time T is given by:

$$\mathbb{E}[g(S_T)|S_t] = \int_0^\infty \underbrace{p(K, T|S_t, t)}_{\text{Transition probability}} \times g(K) dK$$

For any payoff $g(S_T)$, the sifting property of a Dirac-delta function implies that:

$$\begin{aligned} Ig(S_T) &= \int_0^\infty g(K) \delta(S_T - K) dK \\ &= \int_0^F g(K) \delta(S_T - K) dK + \int_F^\infty g(K) \delta(S_T - K) dK \end{aligned}$$

In particular, the dirac-delta function has symmetry. $\delta(x) = \delta(-x)$. The Heaviside step function $\theta(x) = \int_{-\infty}^x \delta(y) dy$. The Heaviside function satisfies $\theta(x) = -\theta(-x)$. Thus,

$$\delta(K - S_T) = \frac{\partial}{\partial K} \theta(K - S_T) = -\frac{\partial}{\partial K} \theta(S_T - K) =$$

We integrate each integrand by parts. The first integral simplifies to:

$$\int_0^F g(K) \delta(S_T - K) dK = \left[g(K) \left(\int \delta(S_T - K) dK \right) \right]_{K=0}^{K=F} - \int_0^F g'(K) \left(\int \delta(S_T - K) dK \right) dK$$

The second integral simplifies to:

$$\int_F^\infty g(K)\delta(K - S_T)dK = \left[g(K) \int \delta(S_T - K)dK \right]_{K=F}^{K=\infty} - \int_0^F \left(\int \delta(S_T - K)dK \right) g'(K)dK$$